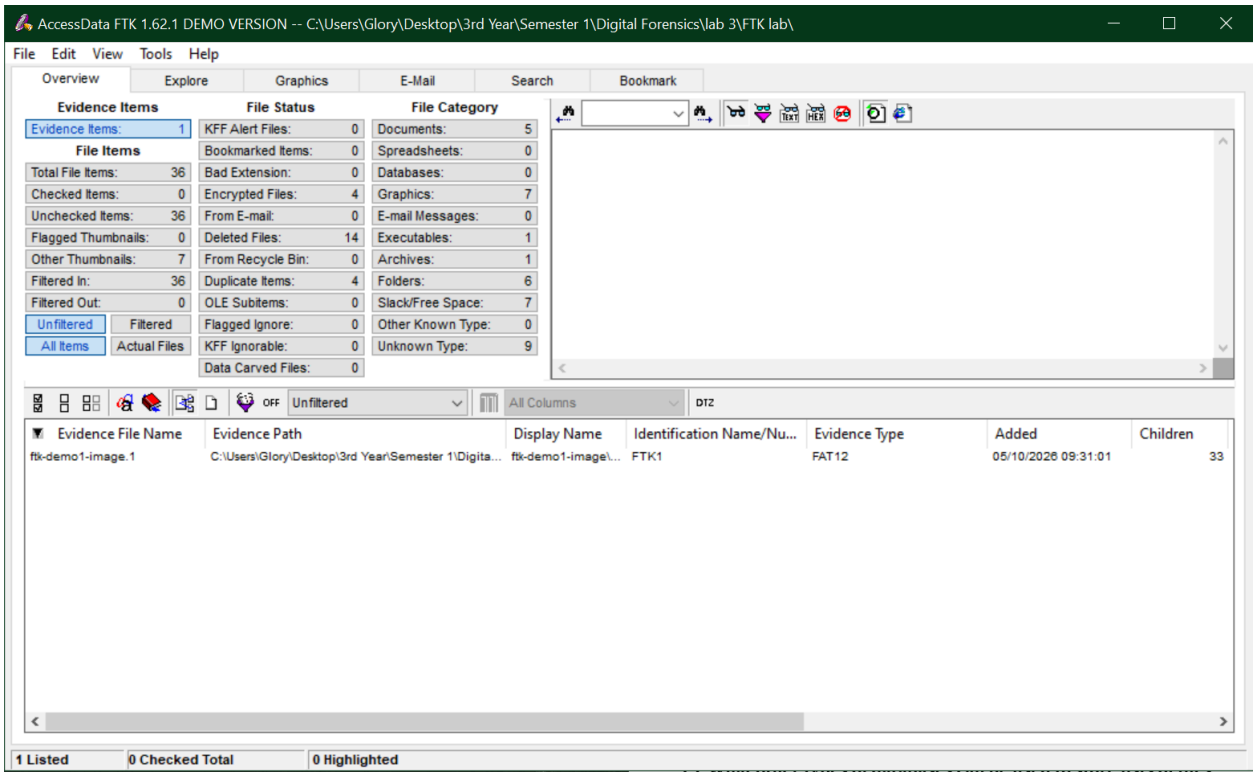
Digital Forensics Lab 3 FTK Investigation

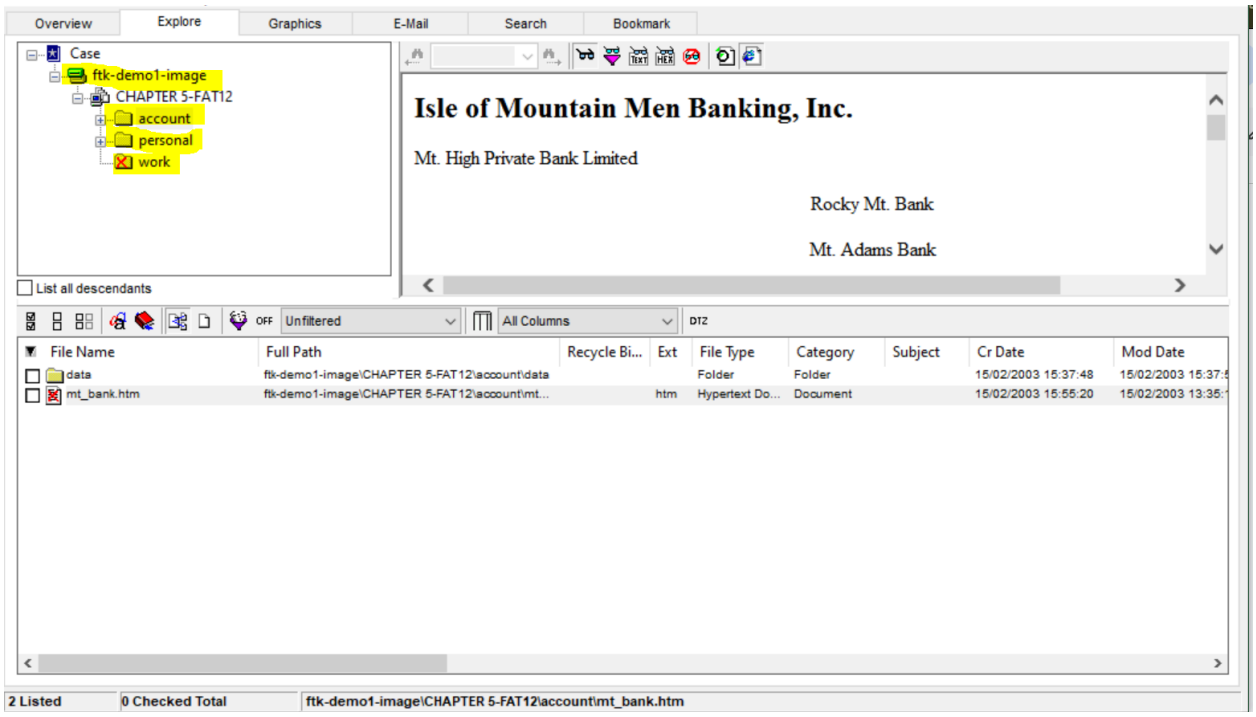
Student	Glory Isoken Oladipo Omoyemense
Module	Digital Forensics

Overview tab



This is the FTK toolkit overview once the configurations are set up and the image is uploaded.

File Layout



This is the file layout for the image file. It shows us an Account folder, Personal Folder and Work folder.

Found Information

1. Texts, Messages and Documents

Names and email addresses

- The recovered emails show the names George Jones and Martha James.
- Their email addresses are georgej@widgets_intl.com and marthaj@widgets_intl.com.
- The original case brief identifies them as George Montgomery and Martha Heiser. It is unclear whether the names in the emails are aliases or whether there is another explanation for the difference.

From: Jones, George [mailto:georgej@widgets_intl.com]
Sent: 29 December 2001 10:52
To: James, Martha [marthaj@widgets_intl.com]
Subject: Re: A plan

I can't talk about it now, it will have to wait until we can talk in private. Maybe Year's Eve.

George

------Original Message------
From: Jones, George [mailto:georgej@widgets_intl.com]
Sent: 26 December 2001 08:02
To: James, Martha [marthaj@widgets_intl.com]
Subject: A plan

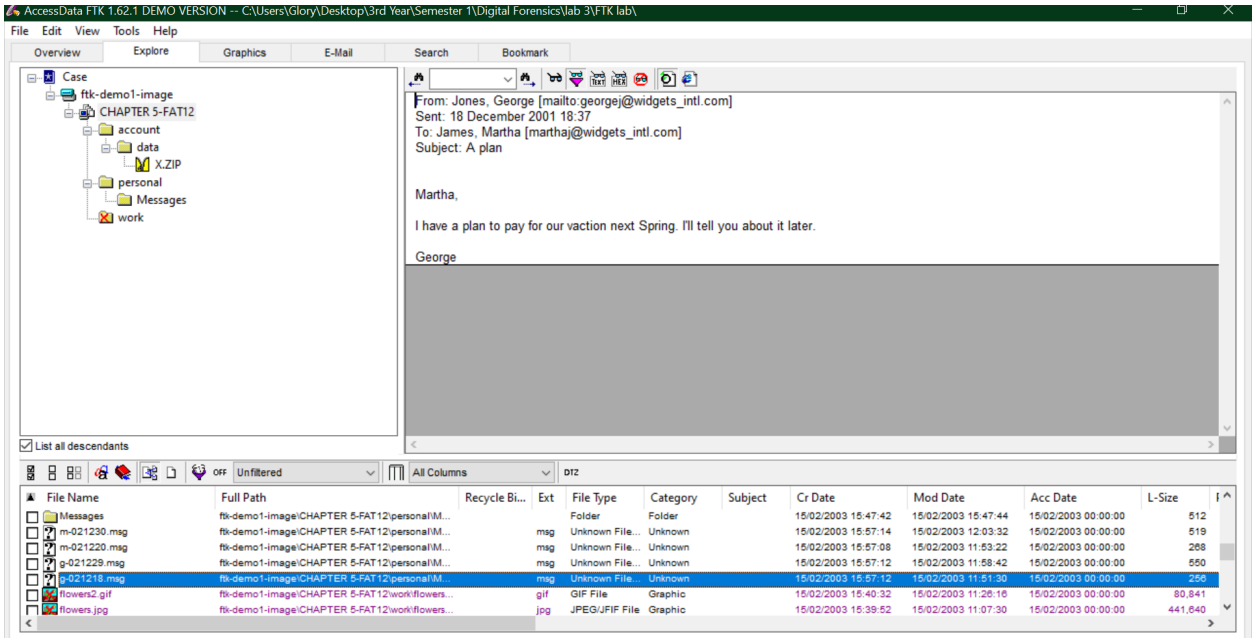
Martha,

I have a plan to pay for our vaction next Spring. I'll tell you about it later.

George

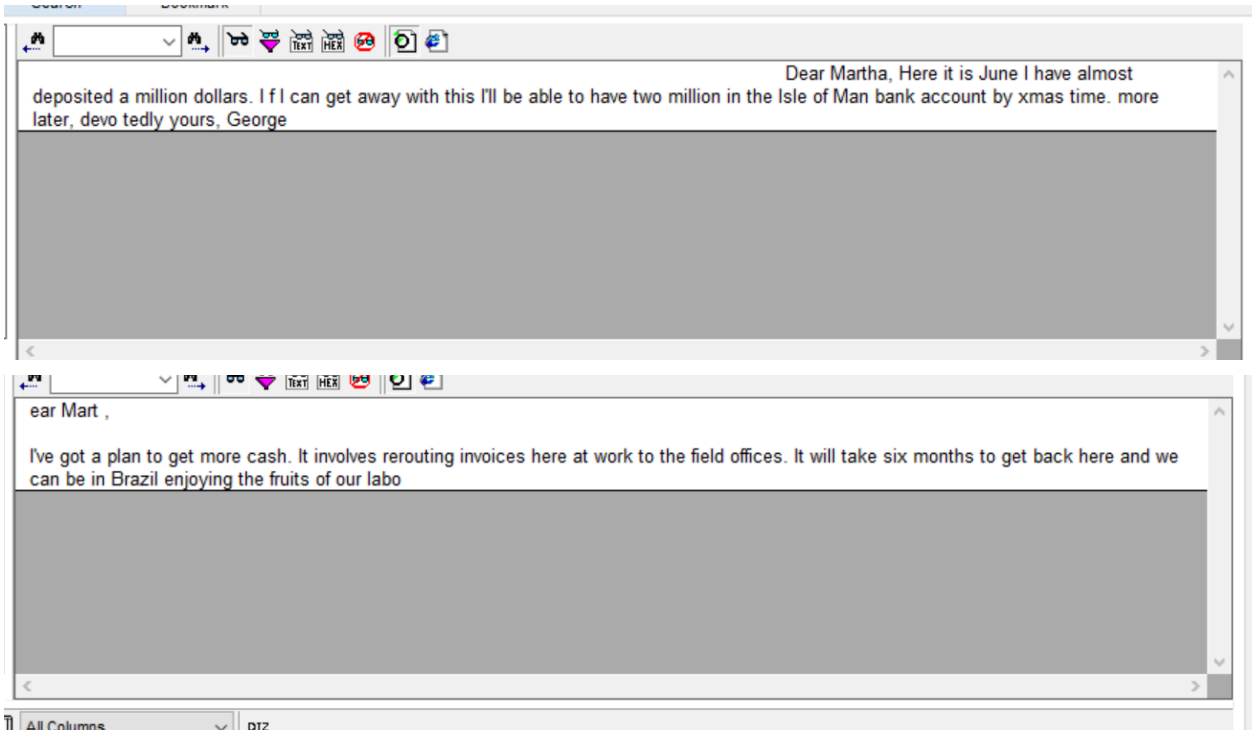
Personal communication

- Several recovered messages suggest George and Martha are romantically involved.
- Their messages discuss future plans, travelling and obtaining more money.



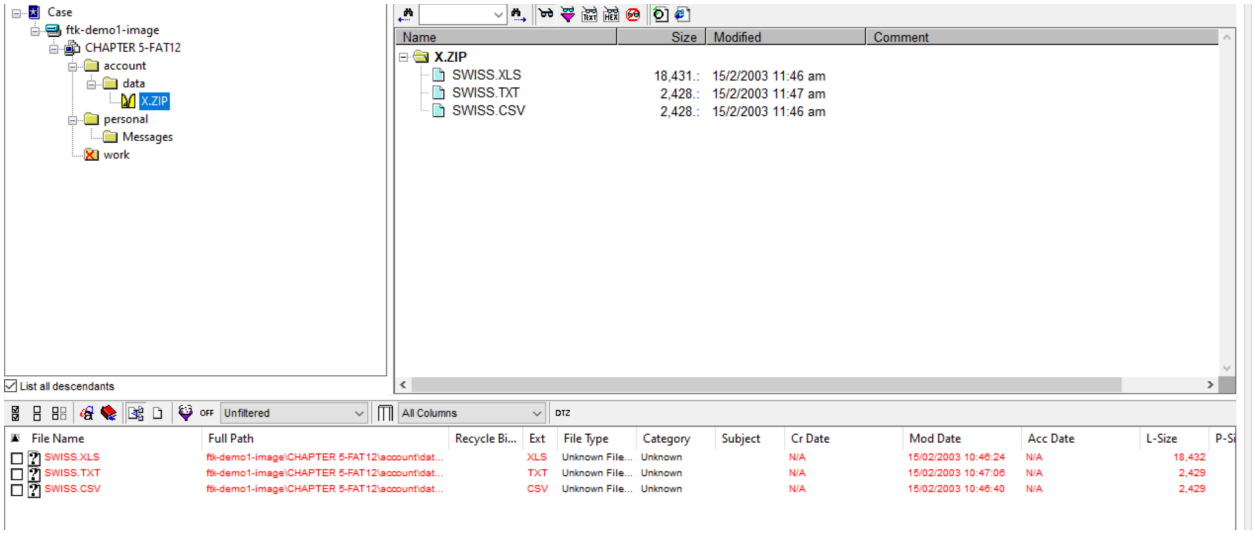
Possible financial misconduct

- George discusses rerouting workplace invoices to field offices as a way of obtaining money.
- In another message, George claims to have deposited $1 million into an Isle of Man bank account, with plans to increase this amount.
- A recovered message mentions missing money, an auditor and concerns about their activities being discovered.



Swiss financial records

- A ZIP archive was found on the evidence drive containing Swiss financial files, including SWISS.CSV and SWISS.TXT.
- Individual copies of these files appeared as deleted in FTK, but the files were still accessible through the ZIP archive.
- The recovered records show deposits and running balances between January 2002 and December 2004.
- The final recorded balance was **$3,898,678.00**, dated 4 December 2004.
- These records may be connected to the suspected invoice fraud, although the source of the money has not been confirmed.



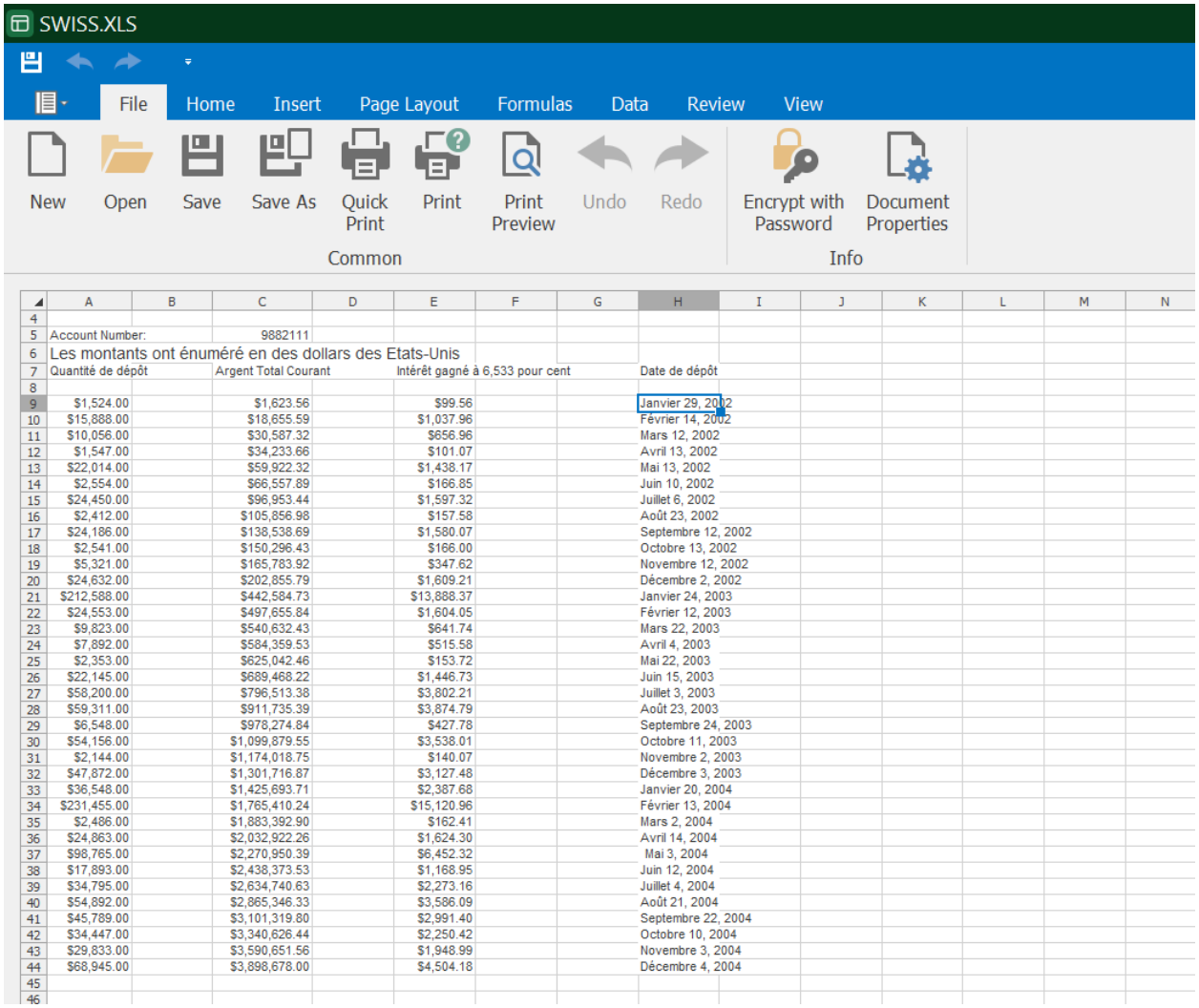
SWISS - Notepad

File Edit Format View Help

Geneve Internationale
Autres liens

Account Number: 9882111
Les montants ont ,num,r, en des dollars des Etats-Unis
Quantit, de d,p‟t Argent Total Courant "Int,rˆt gagn, … 6,533 pour cent" Date de d,p‟t

"$1,524.00"	"$1,623.56"	$99.56	"Janvier 29, 2002"
"$15,888.00"	"$18,655.59"	"$1,037.96"	"F,vrier 14, 2002"
"$10,056.00"	"$30,587.32"	$656.96	"Mars 12, 2002"
"$1,547.00"	"$34,233.66"	$101.07	"Avril 13, 2002"
"$22,014.00"	"$59,922.32"	"$1,438.17"	"Mai 13, 2002"
"$2,554.00"	"$66,557.89"	$166.85	"Juin 10, 2002"
"$24,450.00"	"$96,953.44"	"$1,597.32"	"Juillet 6, 2002"
"$2,412.00"	"$105,856.98"	$157.58	"Ao-t 23, 2002"
"$24,186.00"	"$138,538.69"	"$1,580.07"	"Septembre 12, 2002"
"$2,541.00"	"$150,296.43"	$166.00	"Octobre 13, 2002"
"$5,321.00"	"$165,783.92"	$347.62	"Novembre 12, 2002"
"$24,632.00"	"$202,855.79"	"$1,609.21"	"D,cembre 2, 2002"
"$212,588.00"	"$442,584.73"	"$13,888.37"	"Janvier 24, 2003"
"$24,553.00"	"$497,655.84"	"$1,604.05"	"F,vrier 12, 2003"
"$9,823.00"	"$540,632.43"	$641.74	"Mars 22, 2003"
"$7,892.00"	"$584,359.53"	$515.58	"Avril 4, 2003"
"$2,353.00"	"$625,042.46"	$153.72	"Mai 22, 2003"
"$22,145.00"	"$689,468.22"	"$1,446.73"	"Juin 15, 2003"
"$58,200.00"	"$796,513.38"	"$3,802.21"	"Juillet 3, 2003"
"$59,311.00"	"$911,735.39"	"$3,874.79"	"Ao-t 23, 2003"
"$6,548.00"	"$978,274.84"	$427.78	"Septembre 24, 2003"
"$54,156.00"	"$1,099,879.55"	"$3,538.01"	"Octobre 11, 2003"
"$2,144.00"	"$1,174,018.75"	$140.07	"Novembre 2, 2003"
"$47,872.00"	"$1,301,716.87"	"$3,127.48"	"D,cembre 3, 2003"
"$36,548.00"	"$1,425,693.71"	"$2,387.68"	"Janvier 20, 2004"
"$231,455.00"	"$1,765,410.24"	"$15,120.96"	"F,vrier 13, 2004"
"$2,486.00"	"$1,883,392.90"	$162.41	"Mars 2, 2004"


SWISS.XLS

File Home Insert Page Layout Formulas Data Review View

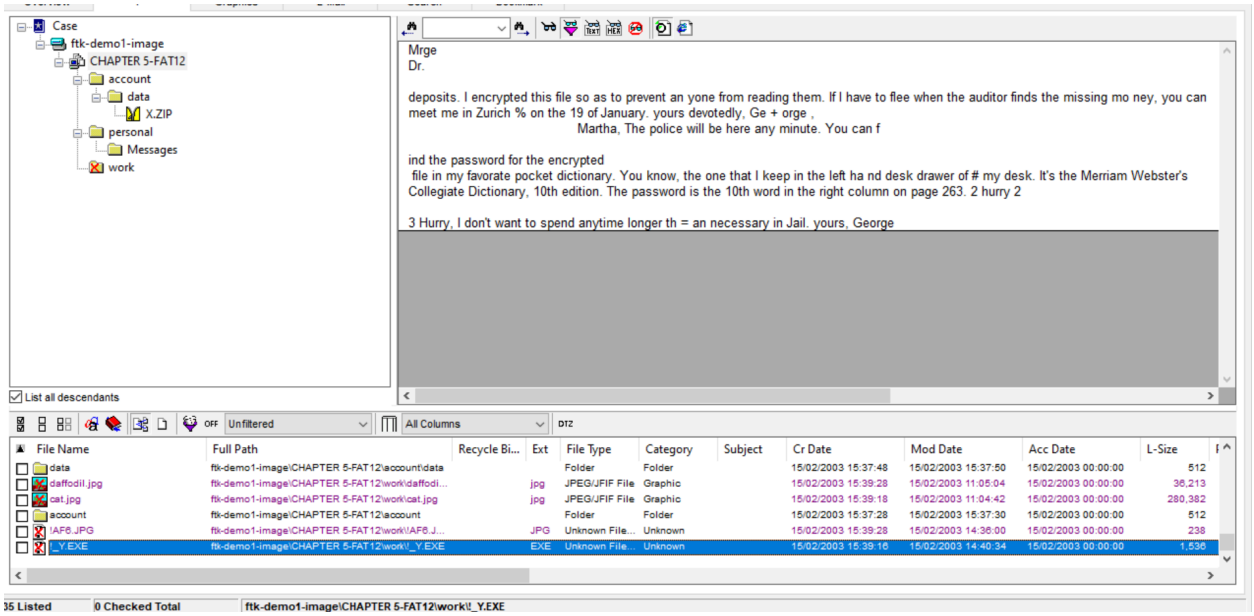
New Open Save Save As Quick Print Print Print Preview Undo Redo Encrypt with Password Document Properties

Common Info

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
4														
5	Account Number:		9882111											
6	Les montants ont énuméré en des dollars des Etats-Unis													
7	Quantité de dépôt		Argent Total Courant		Intérêt gagné à 6,533 pour cent			Date de dépôt						
8														
9	$1,524.00		$1,623.56		$99.56			Janvier 29, 2002						
10	$15,888.00		$18,655.59		$1,037.96			Février 14, 2002						
11	$10,056.00		$30,587.32		$656.96			Mars 12, 2002						
12	$1,547.00		$34,233.66		$101.07			Avril 13, 2002						
13	$22,014.00		$59,922.32		$1,438.17			Mai 13, 2002						
14	$2,554.00		$66,557.89		$166.85			Juin 10, 2002						
15	$24,450.00		$96,953.44		$1,597.32			Juillet 6, 2002						
16	$2,412.00		$105,856.98		$157.58			Août 23, 2002						
17	$24,186.00		$138,538.69		$1,580.07			Septembre 12, 2002						
18	$2,541.00		$150,296.43		$166.00			Octobre 13, 2002						
19	$5,321.00		$165,783.92		$347.62			Novembre 12, 2002						
20	$24,632.00		$202,855.79		$1,609.21			Décembre 2, 2002						
21	$212,588.00		$442,584.73		$13,888.37			Janvier 24, 2003						
22	$24,553.00		$497,655.84		$1,604.05			Février 12, 2003						
23	$9,823.00		$540,632.43		$641.74			Mars 22, 2003						
24	$7,892.00		$584,359.53		$515.58			Avril 4, 2003						
25	$2,353.00		$625,042.46		$153.72			Mai 22, 2003						
26	$22,145.00		$689,468.22		$1,446.73			Juin 15, 2003						
27	$58,200.00		$796,513.38		$3,802.21			Juillet 3, 2003						
28	$59,311.00		$911,735.39		$3,874.79			Août 23, 2003						
29	$6,548.00		$978,274.84		$427.78			Septembre 24, 2003						
30	$54,156.00		$1,099,879.55		$3,538.01			Octobre 11, 2003						
31	$2,144.00		$1,174,018.75		$140.07			Novembre 2, 2003						
32	$47,872.00		$1,301,716.87		$3,127.48			Décembre 3, 2003						
33	$36,548.00		$1,425,693.71		$2,387.68			Janvier 20, 2004						
34	$231,455.00		$1,765,410.24		$15,120.96			Février 13, 2004						
35	$2,486.00		$1,883,392.90		$162.41			Mars 2, 2004						
36	$24,863.00		$2,032,922.26		$1,624.30			Avril 14, 2004						
37	$98,765.00		$2,270,950.39		$6,452.32			Mai 3, 2004						
38	$17,893.00		$2,438,373.53		$1,168.95			Juin 12, 2004						
39	$34,795.00		$2,634,740.63		$2,273.16			Juillet 4, 2004						
40	$54,892.00		$2,865,346.33		$3,586.09			Août 21, 2004						
41	$45,789.00		$3,101,319.80		$2,991.40			Septembre 22, 2004						
42	$34,447.00		$3,340,626.44		$2,250.42			Octobre 10, 2004						
43	$29,833.00		$3,590,651.56		$1,948.99			Novembre 3, 2004						
44	$68,945.00		$3,898,678.00		$4,504.18			Décembre 4, 2004						
45														
46														

Travel plans and search history

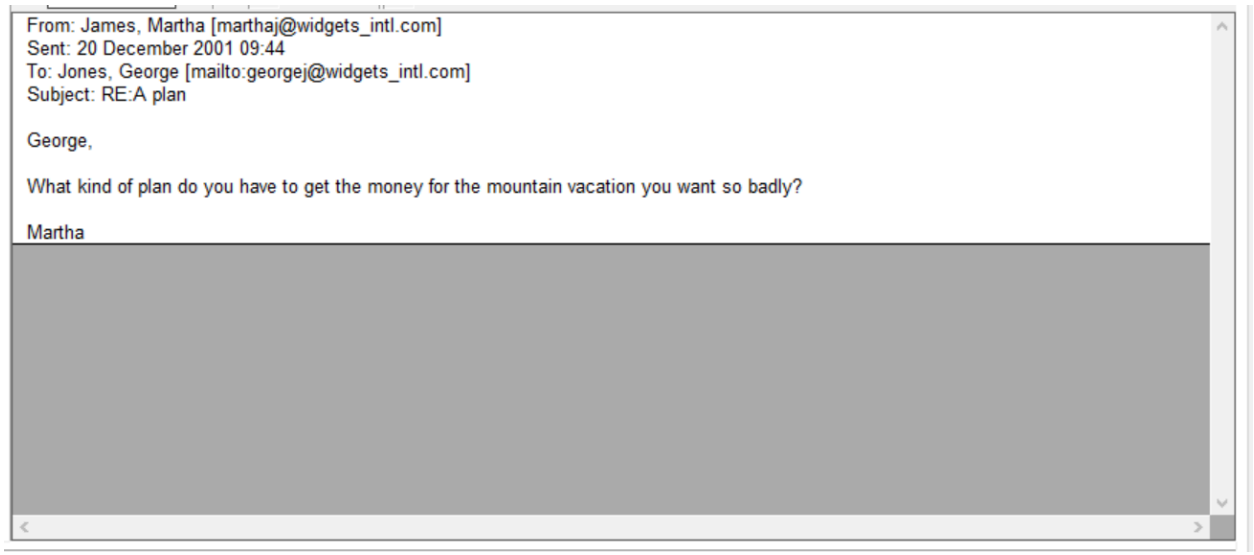
- George and Martha discuss meeting in Zurich, Switzerland, and potentially travelling to Brazil.
- George's browsing history includes searches relating to Geneva, Switzerland.
- His Google Translate history includes French translations, which may be relevant because French is the main language spoken in Geneva.

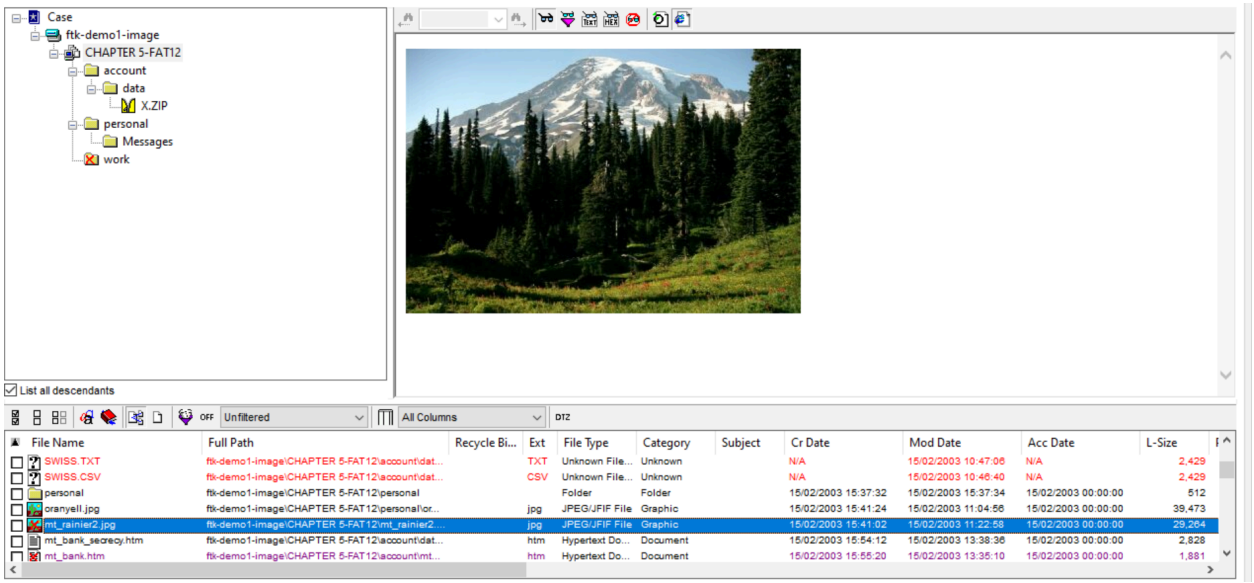


2. Images

Mount Rainier

- An image of Mount Rainier was recovered from the evidence drive.
- This may be relevant because Martha mentions George's desired mountain vacation and questions how he plans to afford it.
- The photograph supports a possible connection to their travel discussions, but does not confirm that the trip took place.





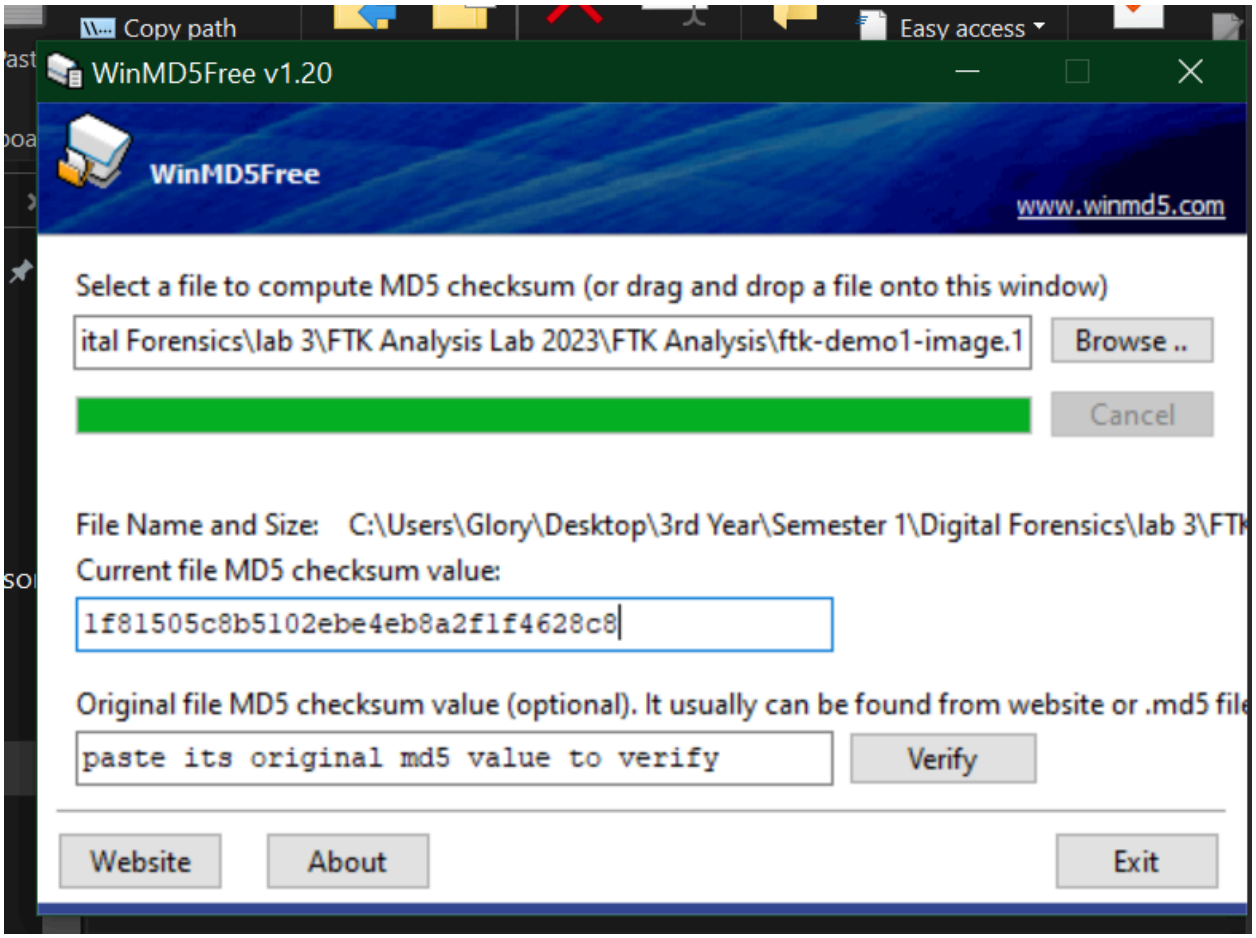
Lab 3 Questions.

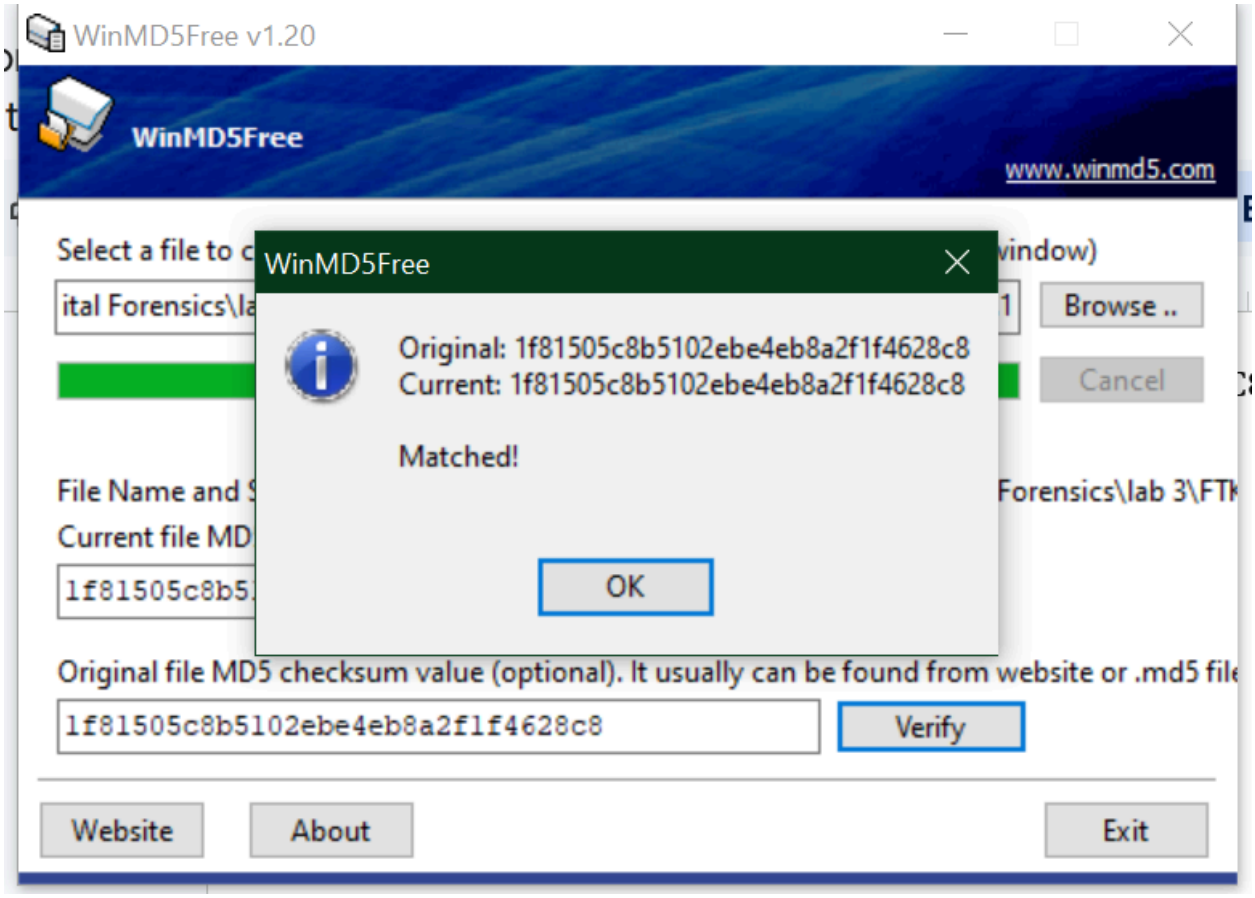
1. **What is the format of the FTK.log file:** The FTK.log file is a text file
2. **What is a log file used for:** It's used for keeping a record of events that occur during the course of a case.
3. **What kind of events can be recorded in a log file:** Case and evidence events, error messages, bookmarking events.
4. **How can you add comments to a log file:** By selecting “Add case Log Entry” under the tools menu item, and then you can look at the file by selecting “View case log ”
5. **List 3 events that can go in a Case Log:** Error messages events, searching events and data carving.
6. **How many bits in a MD5 and SHA1 key? List 3 other Hash Algorithms:** MD5 = 128 bits and SHA1 = 160 bits. 3 other Hash algorithms are, SHA2, SHA3 and bcrypt.
7. **What is the function of the KFF utility:** KFF stands for known file filter. It's a database-matching feature used to find, ignore or flag for specific files on an evidence drive.
8. **What is an Entropy Test:** It is used for known file types to help identify whether the files data is compressed or encrypted.
9. **How can we check if a file is compressed:** You can figure out whether the file is compressed or not by checking the properties of the file and looking at the file system attributes, pile properties and file format, or signature.
10. **Why does data need to be indexed:** Searching, Retrieving and Sorting data becomes much easier when it's indexed.
11. **What does HTML File Listing function do:** It simply creates a HTML version of the file listing.
12. **What is a thumbnail:** a thumbnail is a small preview of an image, video or document.

13. **What other types of databases can be used to store lists of files:** Relational, NoSQL, or embedded databases.
14. **What is data carving? "Research further":** Recovering files or file fragments from raw data using file signatures and structures, even when normal file system information is missing.
15. **What are the options for excluding certain kinds of data:** Filter files by type, size, status or hash values. Known files can also be excluded using KFF.
16. **What is the difference between Slack Space and Free Space? Expand your answer with additional research:** Slack space is unused space within the final allocated cluster of a file. Free space is disk space not currently allocated to active files. Both may contain remnants of deleted data.
17. **What utility compares file hashes against a reference database to eliminate known files:** KFF (Known File Filter).
18. **What is the difference between File Status and File Type:** The difference between file type and status is that, File type tells you the kind of file/format you have and its extension but the file status lets us know its condition in the file system (active/deleted)

Given MD5: 1F81505C8B5102EBE4EB8A2F1F4628C8

Verified MD5: 1F81505C8B5102EBE4EB8A2F1F4628C8





The MD5 checksum of the forensic image was verified using WinMD5Free. The calculated checksum matched the reference value provided, confirming the integrity of the evidence image.

Investigation Overview

The purpose of this lab was to examine a forensic disk image using AccessData FTK and identify evidence relevant to the investigation involving two missing employees.

Investigation Steps

1. I followed the lab instructions to create a new case in AccessData FTK and load the provided forensic image, ftk-demo1-image.1.
2. I used the Overview and Explore tabs to examine the evidence and navigate through the Account, Personal and Work folders.
3. I examined the recovered emails and messages, comparing their contents with the information and hints provided in the lab brief.
4. I investigated references to Geneva and Zurich, including George's browsing and translation history, to understand their possible connection to the travel plans.

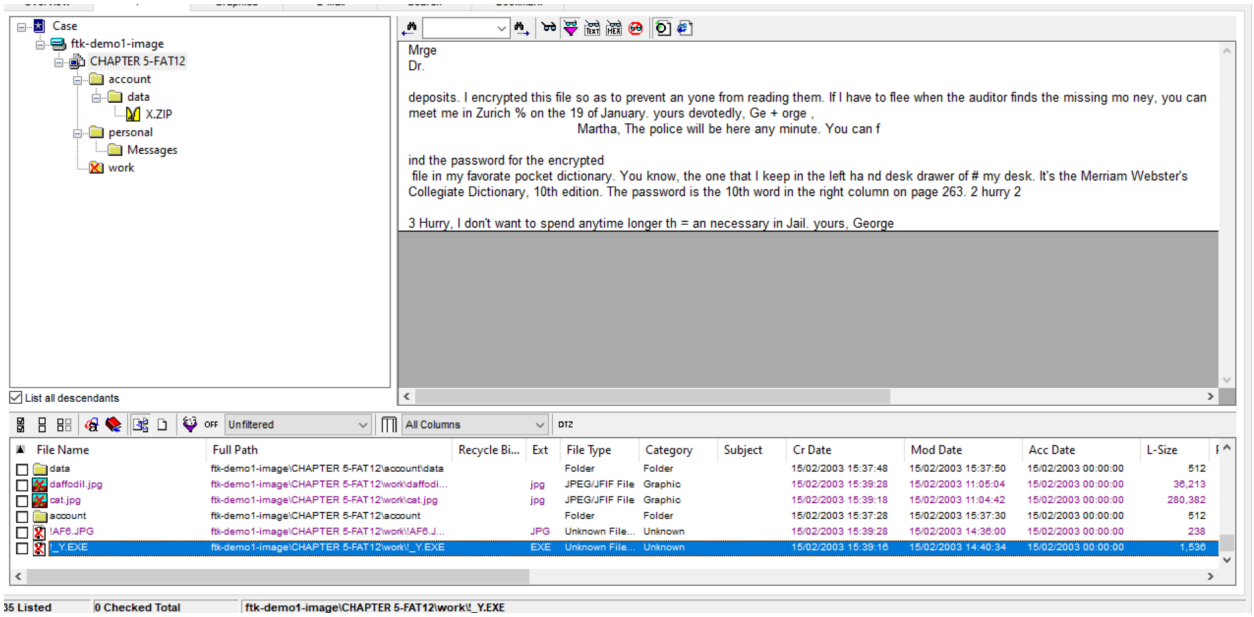
5. I examined the financial documents and attempted to export the individual Swiss CSV, TXT and Excel files. These did not initially open successfully.
6. I located a ZIP archive containing the Swiss files. I exported the archive and was able to open the financial records from it.
7. I examined the recovered financial records and compared them with the messages discussing money, workplace invoices and bank accounts.
8. I reviewed the recovered images and identified a photograph of Mount Rainier, which I connected to a message discussing a mountain vacation.
9. I recorded the relevant findings and screenshots for my investigation report.
10. Finally, I used WinMD5Free to verify the MD5 checksum of the forensic image against the reference checksum provided by the lecturer. Both values matched, confirming the integrity of the image.

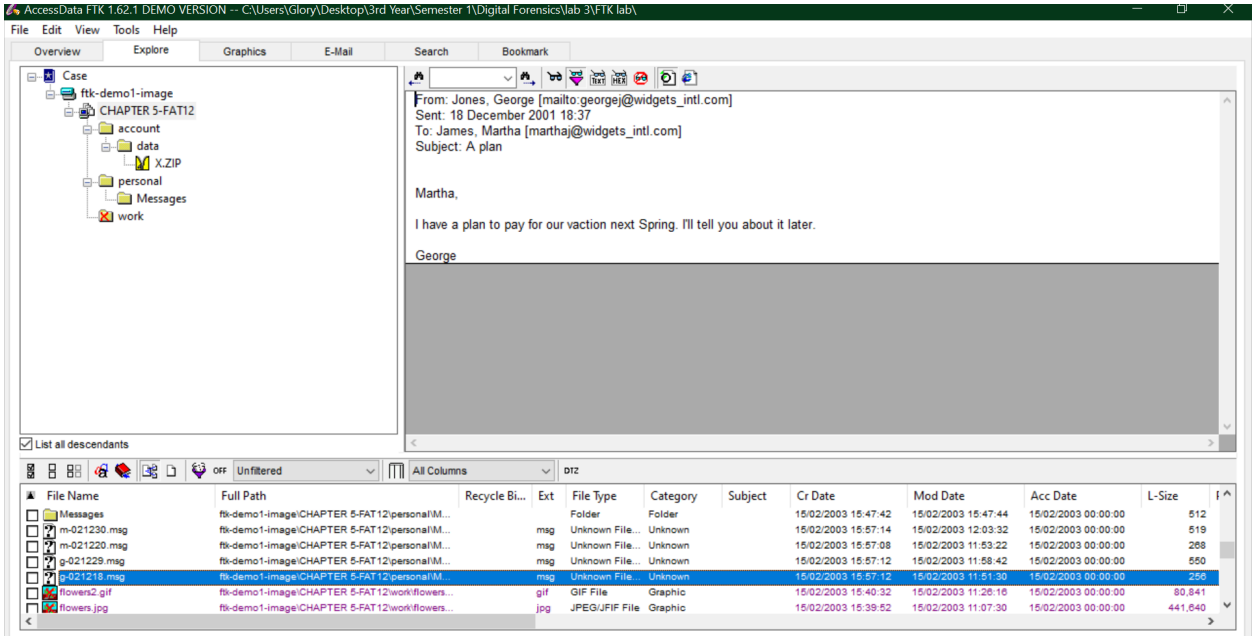
LAB REPORT

Evidence and Findings

1. Emails and communication

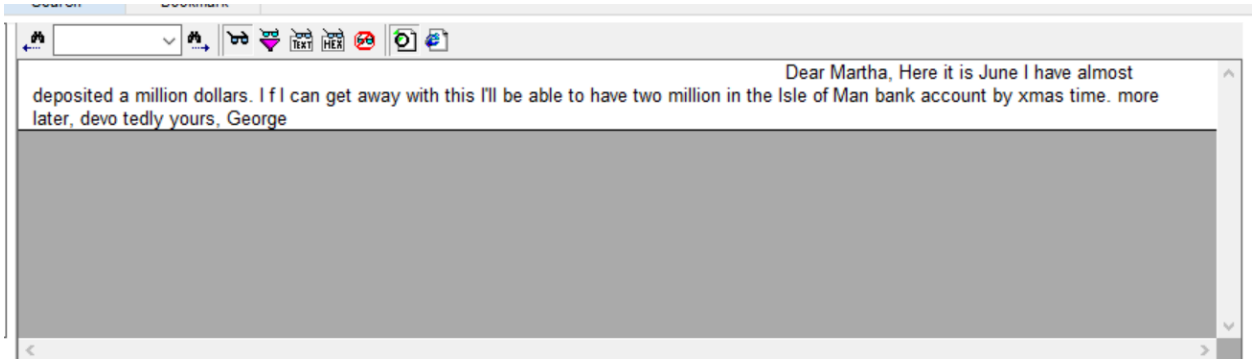
The recovered emails identified individuals named George Jones and Martha James. The messages suggest a personal relationship and contain discussions about obtaining money, rerouting workplace invoices and future travel plans.





2. Possible financial misconduct

One recovered message describes rerouting invoices at work to field offices. Another message claims that one million dollars had been deposited into an Isle of Man bank account. A separate recovered text fragment mentions missing money, an auditor and the possibility of fleeing.



3. Recovered financial records

A ZIP archive contained files including SWISS.CSV and SWISS.TXT. The recovered records were headed *Geneve Internationale* and listed financial deposits and running balances from January 2002 to December 2004.

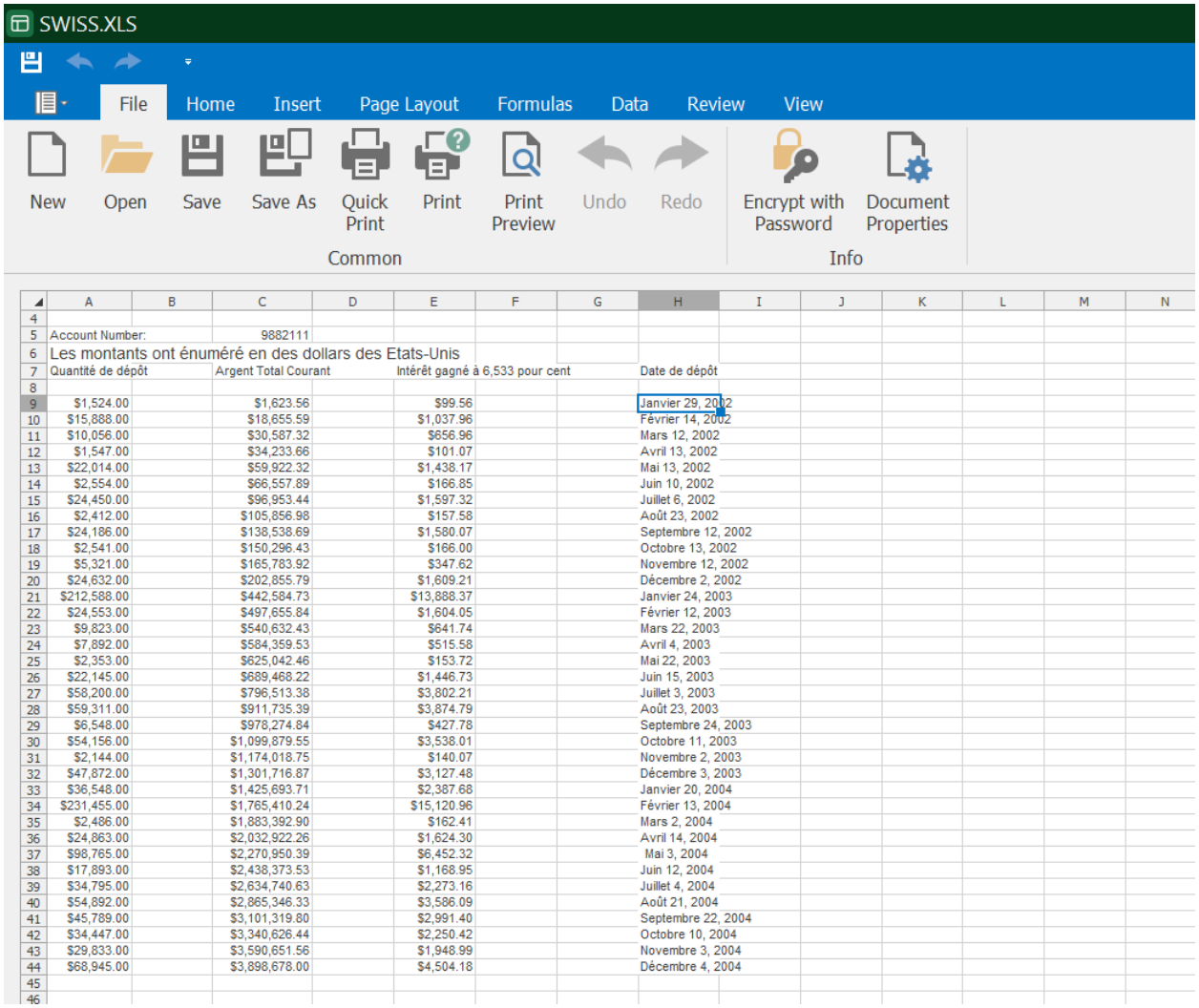
SWISS - Notepad

File Edit Format View Help

Geneve Internationale
Autres liens

Account Number: 9882111
Les montants ont ,num,r, en des dollars des Etats-Unis

Quantit, de d,p‟t	Argent Total Courant	"Int,rˆt gagn, … 6,533 pour cent"	Date de d,p‟t
"$1,524.00"	"$1,623.56"	$99.56	"Janvier 29, 2002"
"$15,888.00"	"$18,655.59"	"$1,037.96"	"F,vrier 14, 2002"
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"$1,547.00"	"$34,233.66"	$101.07	"Avril 13, 2002"
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"$24,186.00"	"$138,538.69"	"$1,580.07"	"Septembre 12, 2002"
"$2,541.00"	"$150,296.43"	$166.00	"Octobre 13, 2002"
"$5,321.00"	"$165,783.92"	$347.62	"Novembre 12, 2002"
"$24,632.00"	"$202,855.79"	"$1,609.21"	"D,cembre 2, 2002"
"$212,588.00"	"$442,584.73"	"$13,888.37"	"Janvier 24, 2003"
"$24,553.00"	"$497,655.84"	"$1,604.05"	"F,vrier 12, 2003"
"$9,823.00"	"$540,632.43"	$641.74	"Mars 22, 2003"
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"$58,200.00"	"$796,513.38"	"$3,802.21"	"Juillet 3, 2003"
"$59,311.00"	"$911,735.39"	"$3,874.79"	"Ao-t 23, 2003"
"$6,548.00"	"$978,274.84"	$427.78	"Septembre 24, 2003"
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"$2,144.00"	"$1,174,018.75"	$140.07	"Novembre 2, 2003"
"$47,872.00"	"$1,301,716.87"	"$3,127.48"	"D,cembre 3, 2003"
"$36,548.00"	"$1,425,693.71"	"$2,387.68"	"Janvier 20, 2004"
"$231,455.00"	"$1,765,410.24"	"$15,120.96"	"F,vrier 13, 2004"
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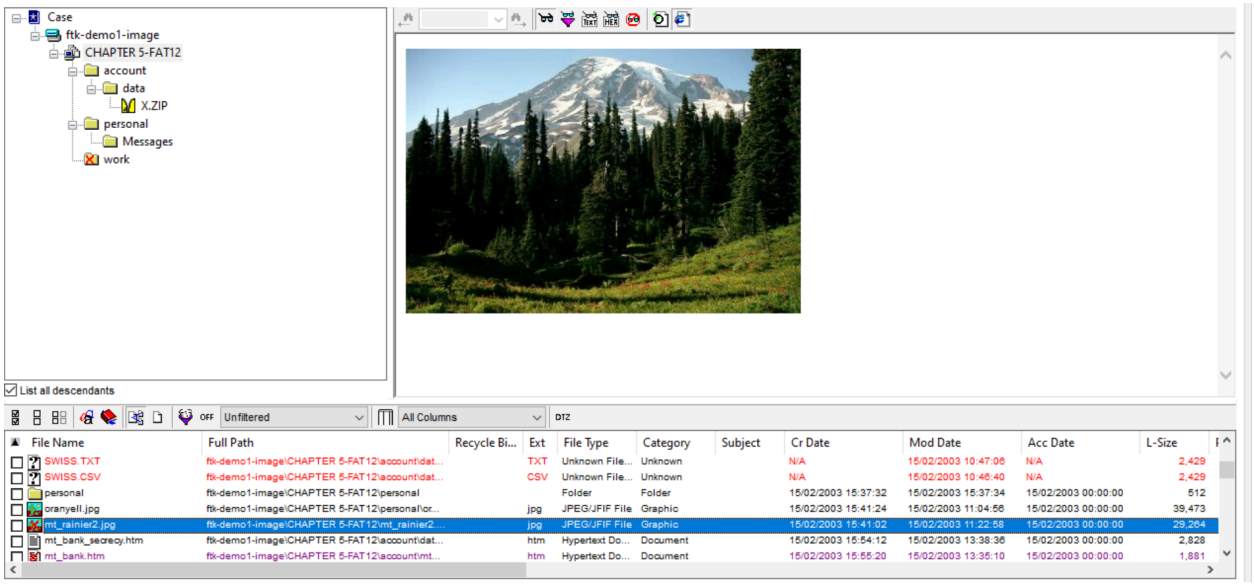
	A	B	C	D	E	F	G	H	I	J	K	L	M	N
4														
5	Account Number:		9882111											
6	Les montants ont énuméré en des dollars des Etats-Unis													
7	Quantité de dépôt		Argent Total Courant		Intérêt gagné à 6,533 pour cent			Date de dépôt						
8														
9	$1,524.00		$1,623.56		$99.56			Janvier 29, 2002						
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15	$24,450.00		$96,953.44		$1,597.32			Juillet 6, 2002						
16	$2,412.00		$105,856.98		$157.58			Août 23, 2002						
17	$24,186.00		$138,538.69		$1,580.07			Septembre 12, 2002						
18	$2,541.00		$150,296.43		$166.00			Octobre 13, 2002						
19	$5,321.00		$165,783.92		$347.62			Novembre 12, 2002						
20	$24,632.00		$202,855.79		$1,609.21			Décembre 2, 2002						
21	$212,588.00		$442,584.73		$13,888.37			Janvier 24, 2003						
22	$24,553.00		$497,655.84		$1,604.05			Février 12, 2003						
23	$9,823.00		$540,632.43		$641.74			Mars 22, 2003						
24	$7,892.00		$584,359.53		$515.58			Avril 4, 2003						
25	$2,353.00		$625,042.46		$153.72			Mai 22, 2003						
26	$22,145.00		$689,468.22		$1,446.73			Juin 15, 2003						
27	$58,200.00		$796,513.38		$3,802.21			Juillet 3, 2003						
28	$59,311.00		$911,735.39		$3,874.79			Août 23, 2003						
29	$6,548.00		$978,274.84		$427.78			Septembre 24, 2003						
30	$54,156.00		$1,099,879.55		$3,538.01			Octobre 11, 2003						
31	$2,144.00		$1,174,018.75		$140.07			Novembre 2, 2003						
32	$47,872.00		$1,301,716.87		$3,127.48			Décembre 3, 2003						
33	$36,548.00		$1,425,693.71		$2,387.68			Janvier 20, 2004						
34	$231,455.00		$1,765,410.24		$15,120.96			Février 13, 2004						
35	$2,486.00		$1,883,392.90		$162.41			Mars 2, 2004						
36	$24,863.00		$2,032,922.26		$1,624.30			Avril 14, 2004						
37	$98,765.00		$2,270,950.39		$6,452.32			Mai 3, 2004						
38	$17,893.00		$2,438,373.53		$1,168.95			Juin 12, 2004						
39	$34,795.00		$2,634,740.63		$2,273.16			Juillet 4, 2004						
40	$54,892.00		$2,865,346.33		$3,586.09			Août 21, 2004						
41	$45,789.00		$3,101,319.80		$2,991.40			Septembre 22, 2004						
42	$34,447.00		$3,340,626.44		$2,250.42			Octobre 10, 2004						
43	$29,833.00		$3,590,651.56		$1,948.99			Novembre 3, 2004						
44	$68,945.00		$3,898,678.00		$4,504.18			Décembre 4, 2004						
45														
46														

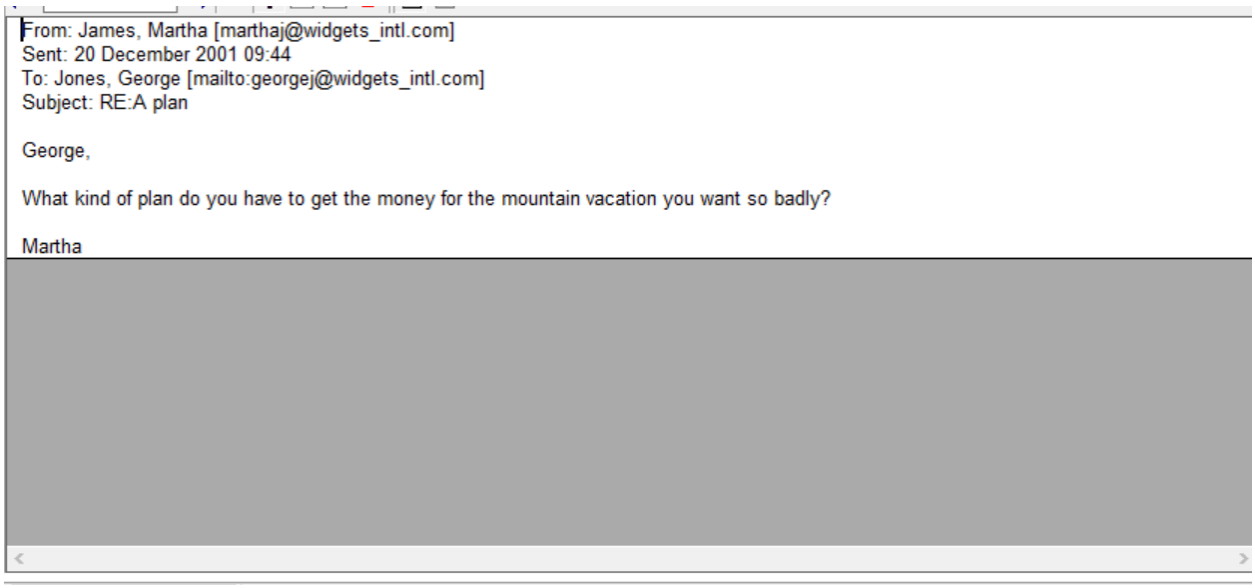
The final recorded running balance was **$3,898,678.00** on 4 December 2004.

4. Travel-related evidence

Recovered messages mentioned Zurich, Switzerland, and possible plans to travel to Brazil. Browser activity also included searches relating to Geneva and French translations.

A photograph of Mount Rainier was found, which may relate to an email discussing a planned mountain vacation.





5. Other files

The evidence image also contained photographs and messages that did not appear directly relevant to the suspected financial activity.

Conclusion

The examination identified communications and financial records suggesting possible workplace financial misconduct. The emails, encrypted-file references and recovered Swiss account records provide several related investigative leads.

However, the evidence examined does not independently confirm the ownership or source of all funds, whether the planned travel occurred, or the reason for the discrepancy between the names in the case brief and recovered emails. Further investigation would be needed to confirm these details.